

Table IV.1: Complete Experimental Environment Specification

Area	Detail
CPU	Intel Core 8th Generation — Family 6, Model 142, Stepping 12 (Whiskey Lake, x86-64)
RAM	16 GB (15 GB available to OS)
Operating System	Windows 11, Build 26200
Python Version	3.13.2
Cryptography	46.0.5 ECDH P-256 and RSA-2048 via hazmat primitives
kybercffi	1.0.1 Kyber512 (ML-KEM-512) key encapsulation
pycryptodome	3.23.0 AES-CBC encryption and PKCS7 padding
hashlib	Python standard library SHA-256, SHA3-256, SHAKE256
numpy / scipy / matplotlib	2.4.2 / 1.17.1 / 3.10.8 statistical analysis and visualisation
ML-KEM Variant	Kyber512 (ML-KEM-512, NIST Security Level 1, ~128-bit post-quantum security) [6][17]
AES Mode	CBC (Cipher Block Chaining) consistent across Scenarios 3 and 4
Padding	PKCS7 via Crypto.Util.Padding (pad / unpad)
Randomness	Keys: library CSPRNG (OS-seeded); IV: os.urandom(16); Plaintext (Hybrid/AES): os.urandom(size); Hash input: deterministic b'A' × size
Timing Function	time.perf_counter() wall-clock, sub-microsecond resolution, returns float seconds
Warm-up Iterations	50 iterations executed and discarded before recording begins (Scenarios 1 and 4)
System Controls	Native hardware (no virtualisation); background applications minimised; power plan set to High Performance to prevent CPU frequency scaling